

SHIVAM SHARMA

Aspiring Software / Web Developer | CSE Undergraduate

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SUMMARY

CSE undergraduate and aspiring software/web developer skilled in building responsive, cross-platform web applications with HTML5, CSS3, JavaScript, and React.js. Comfortable across the stack — from UI design in Figma to backend logic and databases (MySQL) with additional experience in Python-based data analysis and visualization (NumPy, Pandas, Matplotlib, Seaborn). Strong foundation in data structures, algorithms, and problem-solving; eager to contribute to real-world engineering teams.

TECHNICAL SKILLS

Languages & Markup: JavaScript, TypeScript, Python, C/C++, Java, SQL, HTML5, CSS3

Frameworks & Libraries: React.js, Next.js, Tailwind CSS, Framer Motion, Mongoose

Databases & Backend Services: MongoDB, PostgreSQL, Pinecone, Firebase

Cloud & Developer Tools: Docker, Git, GitHub, Postman, Render, Vercel, Cloudinary

Data & ML Libraries: Pandas, NumPy, Seaborn, OpenCV, Matplotlib

Core CS: Object-Oriented Programming, DBMS, Operating Systems, Data Structures, Software Engineering, REST APIs

AI / ML: RAG, LangChain, LangGraph, Prompt Engineering

INTERNSHIP EXPERIENCE

Data Analysis Intern — CERN, *European Organization for Nuclear Research*

Aug 2026

- Worked with CERN ROOT, a data analysis framework widely used in high-energy and nuclear physics research.
- Analysed and processed nuclear physics experimental data using ROOT.
- Created and customized histograms, graphs, and data visualizations to study experimental results.
- Used ROOT for data fitting and statistical analysis to extract meaningful physical parameters.
- Developed basic C++ scripts/macros for automating data analysis and visualization tasks.
- Worked with ROOT files, TTrees, and branches to access and organize large experimental datasets.
- Improved skills in C++, data analysis, plotting, error analysis, and scientific problem-solving through hands-on research work.

Software Engineer Intern — IEPSSOLS & Co.

Apr 2026 – Jul 2026

- Developed Modular Android UI:** Built reusable, high-performance UI components using Kotlin and Jetpack Compose (or XML) to standardize app layouts and speed up feature delivery.
- Diagnosed & Fixed Critical Bugs:** Leveraged Android Studio Profiler and Logcat to debug complex runtime crashes and edge-case defects, significantly improving app stability.
- Collaborated on End-to-End Features:** Partnered with senior engineers to design, test, and seamlessly integrate backend APIs into core mobile application modules.

Data Visualization — Forage Virtual Experience Program

Mar 2025 – Aug 2025

- Completed a job-simulation program analyzing business scenarios to identify key data questions.
- Selected and built effective visualizations to communicate complex data clearly.
- Practiced data storytelling by turning raw data into a clear visual dashboard.
- Cleaned and prepared large transactional datasets in Power BI by filtering out negative quantities and price errors.

PROJECTS

Personal Portfolio Website

HTML5, CSS3, JavaScript

- Designed and built a responsive personal website to showcase projects and technical case studies, optimized for cross-device compatibility.
- Engineered a custom, modern UI without a framework, focusing on performance and clean semantic markup.
- Live:** shivamsharmadiatm.netlify.app

Crop Prediction Using AI

GEN AI, MERN STACK

- Built an AI-based system to predict suitable crop type and expected yield from environmental and soil parameters.
- Aimed at improving agricultural productivity and reducing crop failure risk through data-driven recommendations.
- Live:** ai-crop-prediction.onrender.com

HTTP Web Server

C++, HTTP Protocol, Non-Blocking I/O, CGI

- Developed an HTTP server in C++ capable of handling GET, POST, and DELETE methods with support for cookies, CGI, file uploads/downloads and static content delivery.
- Implemented non-blocking I/O operations to support simultaneous reading and writing with multiple clients connected.
- Handled HTTP requests, headers, status codes, client connections, and server responses according to HTTP protocol requirements.

- Strengthened understanding of network programming, TCP/IP, HTTP protocols, concurrency, and low-level C++ system programming.

- **Live:**

OutreachX

Next.js, Node.js, Firebase, LangChain, OpenRouter, VAPI, Twilio, Pinecone

- Built an agentic AI campaign automation platform with multi-step workflow orchestration, supporting WhatsApp, voice notes, and AI phone calls through a 6-step no-code campaign builder.
- Implemented RAG-based document-grounded replies using Pinecone vector search; surfaced low-confidence and unanswerable queries on a live dashboard to identify knowledge gaps.
- Built a real-time analytics dashboard to track delivery metrics, campaign engagement, and channel-wise outreach performance.

- **Live:**

Avento

Next.js 15, MongoDB Atlas, Pinecone, OpenRouter, Clerk

- Developed a RAG-powered customer support SaaS enabling businesses to deploy context-aware chatbot widgets on Framer and custom websites in under 60 seconds.
- Engineered document ingestion pipelines with 500-character chunks, 100-character overlap, and 768-dimensional Pinecone embeddings, achieving under 100 ms retrieval latency.
- Built chatbot analytics, knowledge-gap detection, origin whitelisting, branding controls, and an integrated conversational AI calling workflow for context-aware voice-based customer support.

- **Live:**

Perception

Next.js, Fast API, MongoDB, Apify, LangChain

- Built a multilingual social sentiment intelligence platform for brand monitoring, ingesting social media mentions through Apify pipelines and classifying posts using an LLM-powered CX triage engine.
- Developed Genie, a conversational analytics agent that lets analysts query brand sentiment in natural language, generate chart-backed insights, and trigger email reports from the dashboard.
- Architected a Fast API and MongoDB backend to store, aggregate, and surface brand-level and category-level sentiment trends through a live Next.js dashboard.

- **Live:**

Gatherly

Next.js, React, TypeScript, MongoDB, Mongoose, Clerk, Nodemailer, Zoom OAuth

- Built a full-stack event management platform enabling organizers to create events, manage RSVPs, publish public event pages, and handle attendee approval workflows.
- Integrated Clerk authentication, MongoDB/Mongoose data models, Nodemailer transactional emails, and protected organizer routes with ownership-based access control.
- Implemented Zoom OAuth meeting sync using an extension-based architecture, isolating Zoom side effects from core event CRUD logic for cleaner, maintainable backend workflows.

- **Live:**

EDUCATION

Durgapur Institute of Advanced Technology and Management — B.Tech., CSE

Expected Jul 2027

Rajbandh, Durgapur, West Bengal

- Successfully completed my curriculum as one of the top 5 fastest students out of a cohort of over 100.
- Relevant Coursework: Data Structures & Algorithms, Databases, Computer Systems.

SD DAV Public School — Higher Secondary (XII), CBSE

2023

Jamtara, Jharkhand | 85% | Coursework: Database, Python Programming

ACHIEVEMENTS

- Awarded the Academic Excellence Award for the 2024–25 session for outstanding scholastic performance.
- Selected for a research internship in Nuclear Physics at CERN, gaining hands-on experience with the ROOT framework for scientific data analysis and visualization.

CERTIFICATIONS

- MERN Stack Web Development — Novacode (Aug 2025 – Oct 2025)
- SQL: Basic to Advanced — Udemy (Dec 2024)
- Java Data Structures & Algorithms — Udemy (Dec 2024)